Fossil Evidence for Evolution



Tree Ring Dating
Relative Dating
Stratigraphy
Phylogenetic Trees
Geological Time Scale
Problems with Fossil Record

Tree Ring Dating (Dendrochronology)

- Another form of absolute dating
- Trees grow concentric rings
  - Each ring = 1 year growth
  - Differ in width according to growing season
  - Create a “Timeline” in the tree trunk
  - Can match “years” by comparing one sample to another.
  - Can therefore match historically, and count years.

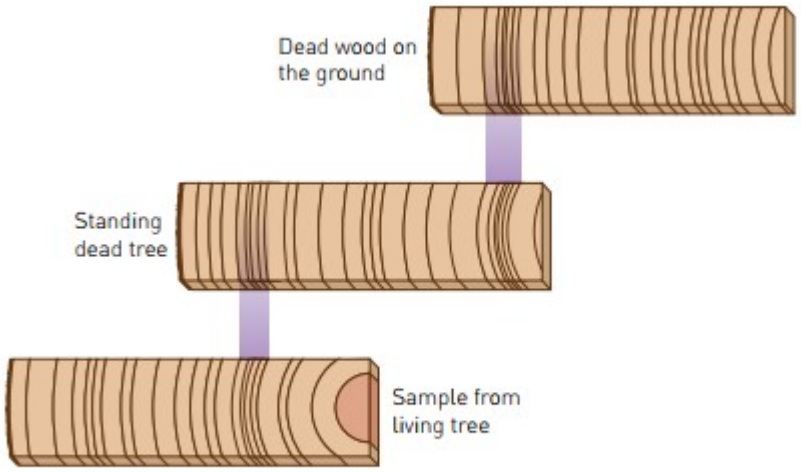


Figure 16.7 Building a tree ring chronology. Overlapped samples from living and dead trees extend the period of dating back to prehistoric times.

Tree Ring Dating (Dendrochronology)

- Old, living trees are particularly useful for this
  - Bristlecone pine – USA – living trees up to 4500 years old
- Can then look at dead trees, and backtrack using comparison
  - Can date very accurately back to 8600 years ago
  - Can use this to verify accuracy of Carbon-14 dating
- Example: Seahenge
  - Timber circle – coast of England
  - Analysis of central stump using combo of tree-ring and carbon dating
  - Middle stump: tree chopped down in 2049 BC (4000 years ago), and was 167 years old when felled.



Figure 16.8 Seahenge

Tree Ring Dating (Dendrochronology)

- Limitations:
  - Timber only rarely preserved long enough to do long dating
  - Depends on having a continuous succession of nearby timber

Table 16.1 Useful range of common absolute dating methods

Dating method	Material used	Useful range (years BP)
Tree growth rings	wood	up to 9000
Carbon-14	carbon compounds	up to 60 000
Protactinium	sea sediments	up to 250 000
Uranium-thorium	sea sediments, coral	up to 600 000
Potassium-argon	volcanic deposits	200 000 and earlier
Electron spin resonance	calcium carbonate (in shells, coral, teeth), also quartz and flint	up to 100 000, possibly 300 000
Fission tracks	minerals and glass	100 years ago to 4550 million
Thermoluminescence	sediments, lava, ceramics	300 years ago to 100 000

Relative Dating - Stratigraphy

- Determines how old one sample is, compared to other samples
- Enables sequence of events to be established
- Stratigraphy: study of layers, or strata
  - Principle of Superposition
    - In sedimentary rock, layers at top are younger than layers beneath
    - Need to be cautious, as distortions of Earth's crust can disturb layers, leading to inaccuracy.
  - Correlation of Rock Strata
    - Matching layers of rock from different areas
    - Examining rock, and also fossils it contains
    - Rocks containing similar fossils are likely to be of similar age

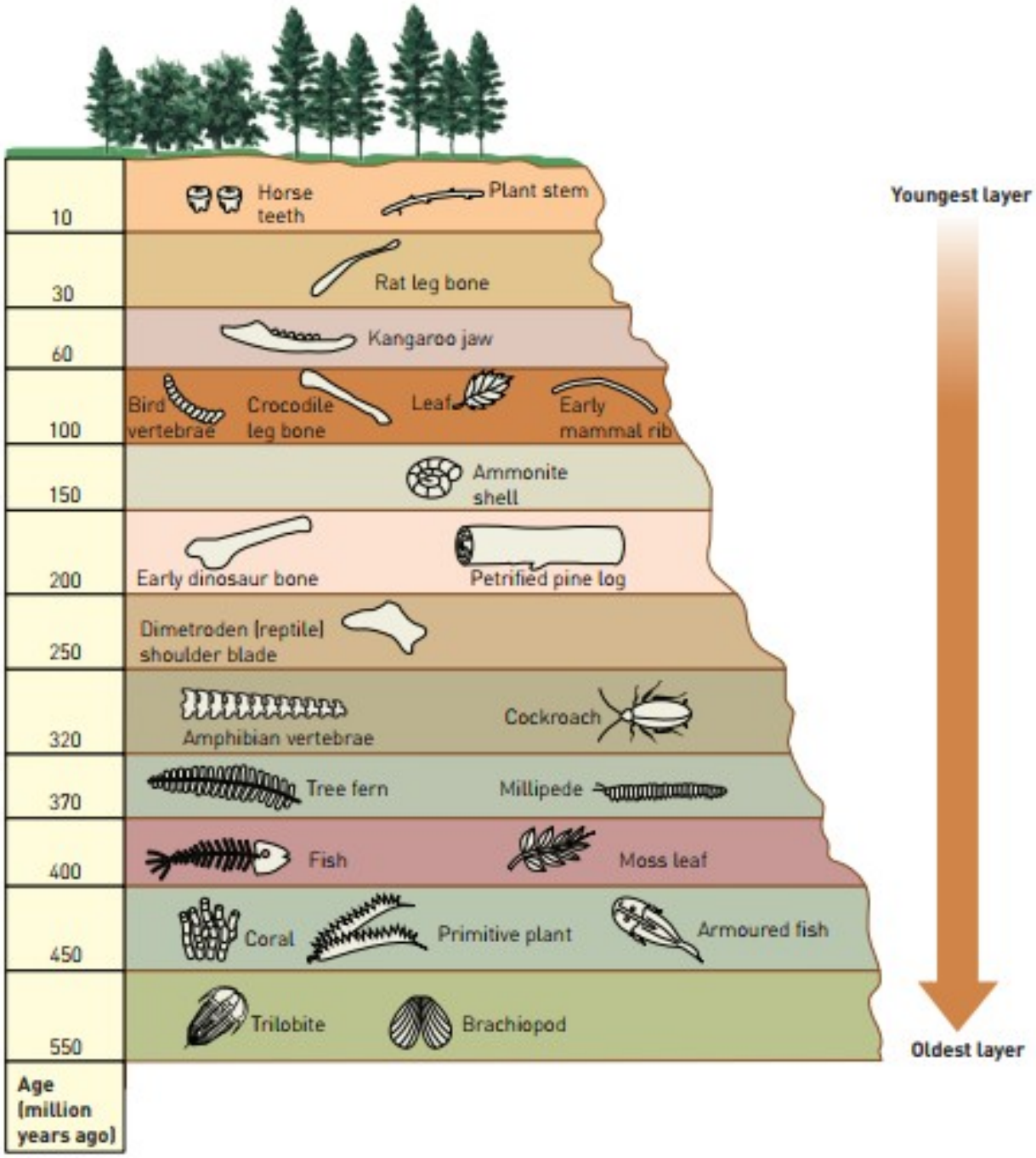
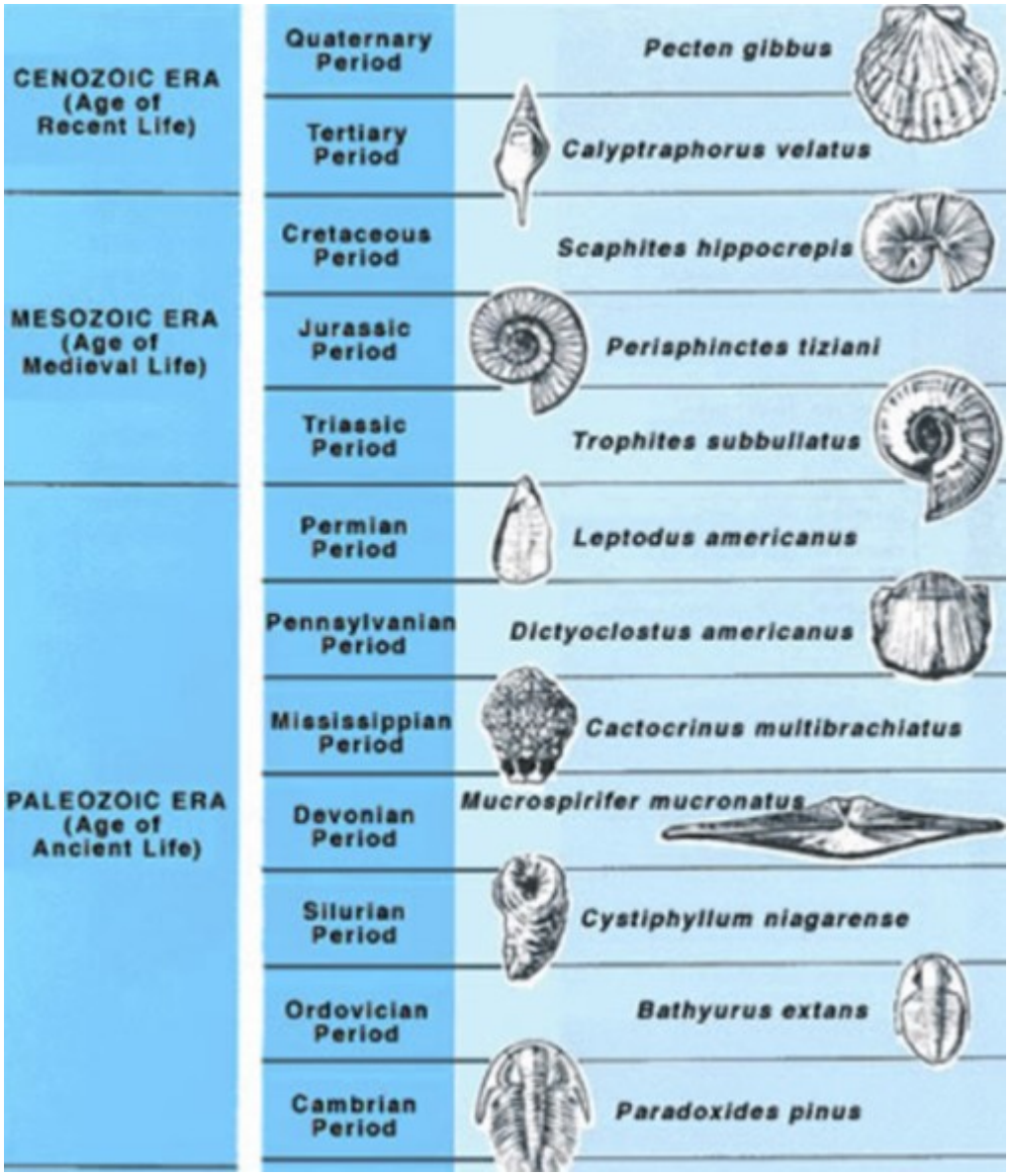


Figure 16.9 A section of rock strata containing fossils of different ages. The principle of superposition assumes that the younger strata are towards the top of the sequence.

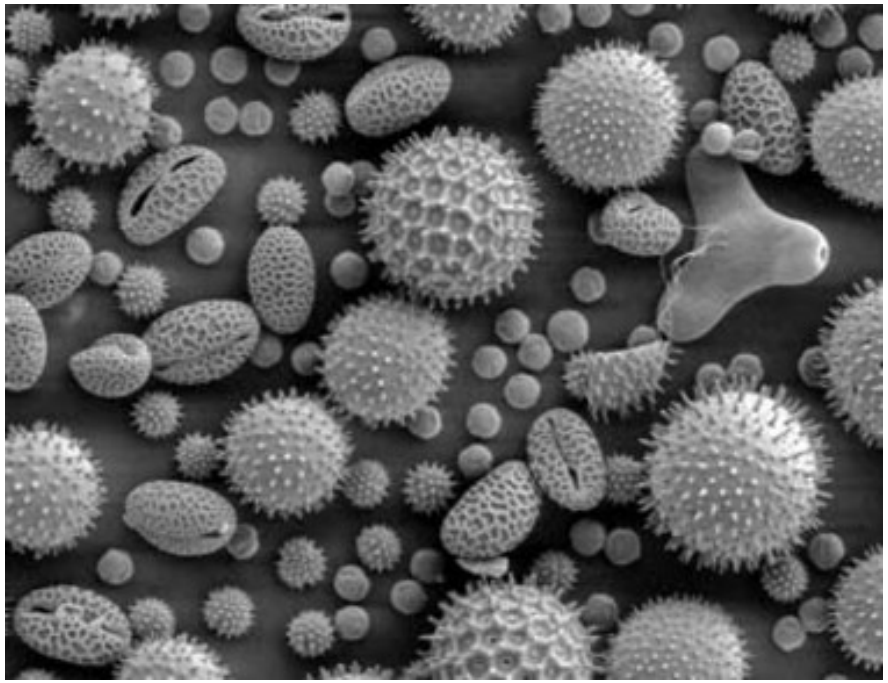
Relative Dating - Stratigraphy

- Index fossils:
  - Widely distributed on earth but only present for a short historical time.
  - Can be used to correlate strata from different areas of the earth



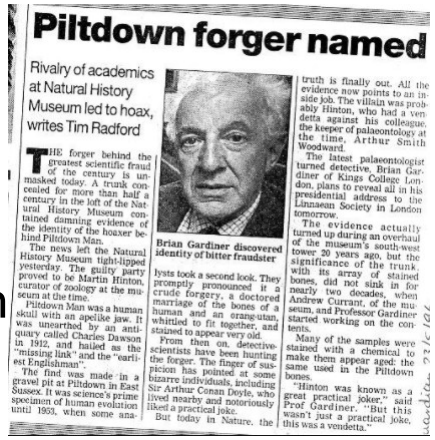
Relative Dating - Stratigraphy

- Fossil pollen grain analysis
  - Some useful as Index Fossils
  - Can be used to get a picture of type and amount of vegetation present at the time the fossils formed.
  - Can be used to infer information about climate
  - Can then be used to confirm relative dates



Relative Dating – Fluorine Dating

- Bone left in soil, fluoride ions replace some of the ions in the bone.
- All fossil bones in a particular deposit should contain same amount of fluoride
  - Can therefore detect fossils that have been displaced
- Older fossils, more fluorine as have been there longer
- Can't use to absolute date though, as amount of fluoride in water supply fluctuates over time.
- Example: "Piltdown Man" hoax
  - 1953 – fossil claimed to be of "human ancestor"
  - Fluorine dating showed it to be a hoax:
    - Skull old, but jawbone modern – from orang-utan



Piltdown forger named

Rivalry of academics at Natural History Museum led to hoax, writes Tim Radford

THE forger behind the greatest scientific fraud of the century is unmasked today. A trunk concealed for more than half a century in the loft of the Natural History Museum contained damning evidence of the identity of the hoaxer behind Piltdown Man.

The news left the Natural History Museum tight-lipped yesterday. The guilty party proved to be Martin Hinton, curator of zoology at the museum at the time.

Piltdown Man was a human skull with an apelike jaw. It was unearthed by an antiquary called Charles Dawson in 1912, and hailed as the "missing link" and the "earliest Englishman".

The find was made in a gravel pit at Piltdown in East Sussex. It was science's prime specimen of human evolution until 1953, when some analysts took a second look. They promptly pronounced it a crude forgery, a doctored marriage of the bones of a human and an orang-utan, whittled to fit together, and stained to appear very old.

From then on, detective-scientists have been hunting the forger. The finger of suspicion has pointed at some bizarre individuals, including Sir Arthur Conan Doyle, who lived nearby and notoriously liked a practical joke.

But today in Nature, the truth is finally out. All the evidence now points to an inside job. The villain was probably Hinton, who had a vendetta against his colleague, the keeper of palaeontology at the time, Arthur Smith Woodward.

The latest palaeontologist turned detective, Brian Gardiner of Kings College London, plans to reveal all in his presidential address to the Linnaean Society in London tomorrow.

The evidence actually turned up during an overhaul of the museum's south-west tower 20 years ago, but the significance of the trunk with its array of stained bones, did not sink in for nearly two decades, when Andrew Currant, of the museum, and Professor Gardiner started working on the contents.

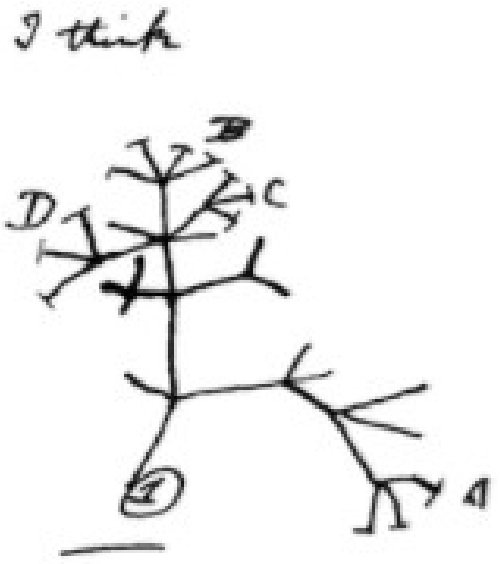
Many of the samples were stained with a chemical to make them appear aged: the same used in the Piltdown bones.

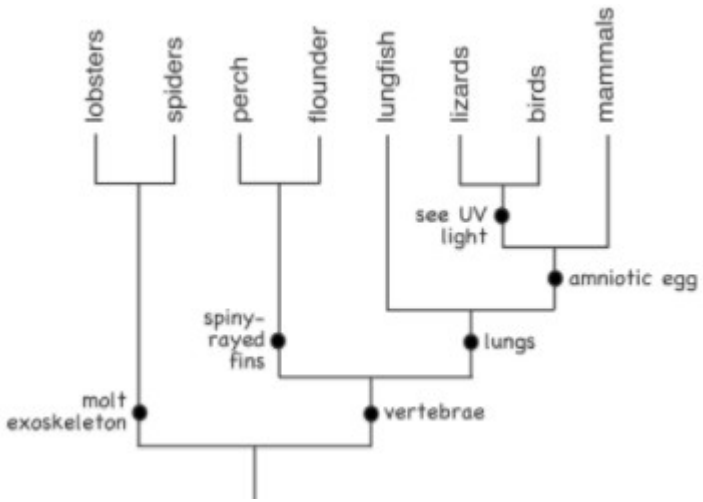
"Hinton was known as a great practical joker," said Prof Gardiner. "But this wasn't just a practical joke, this was a vendetta."

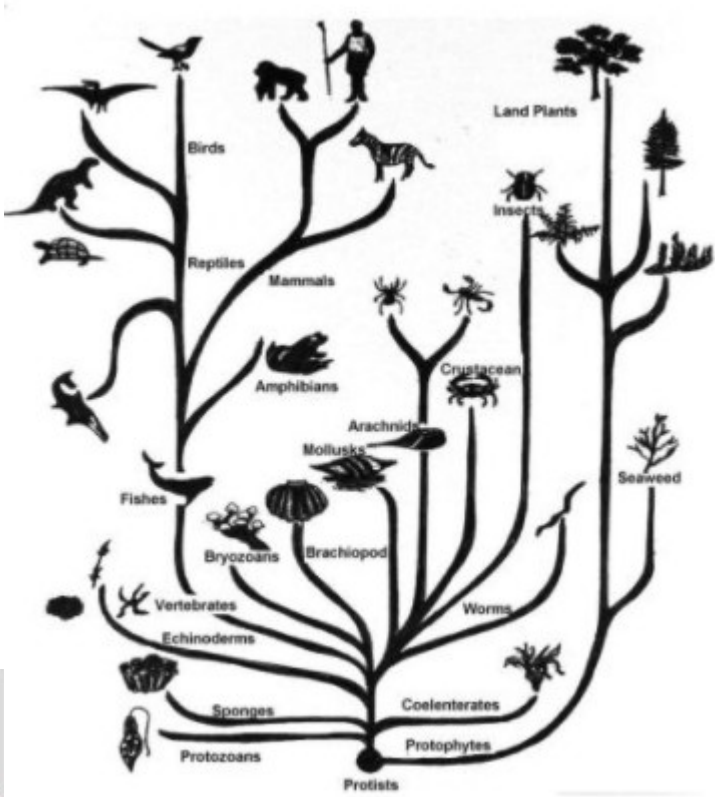
Brian Gardiner discovered identity of bitter fraudster

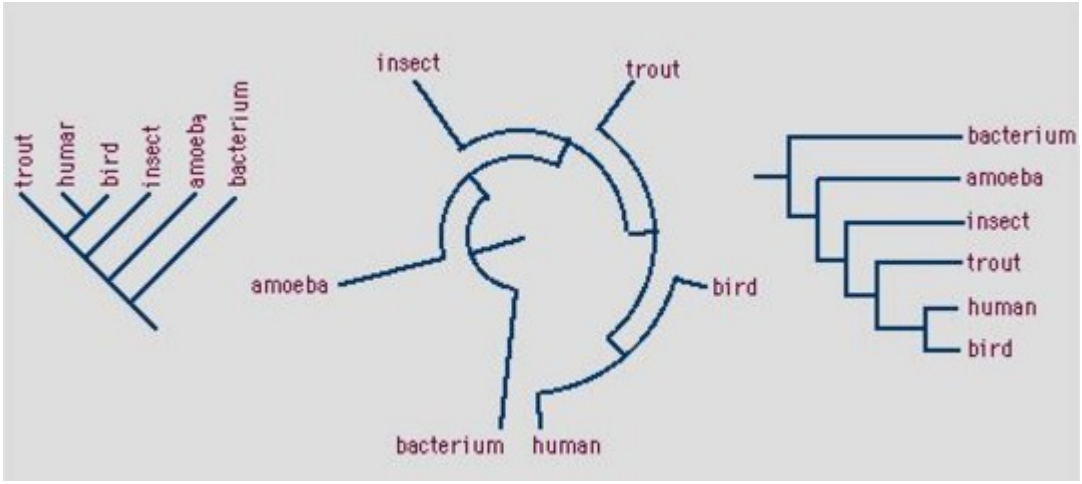
Phylogenetic Trees

- Using a range of dating, biotech and classification techniques, scientists can work out likely evolutionary relationships
- Probably relationships can be represented as a diagram, called a *phylogenetic tree*.
  - Ancestral organism at the base
  - More recent organisms at end of branches
  - Forks in branches: common ancestors
- Have been used since Darwin's time.




lobsters
spiders
perch
flounder
lungfish
lizards
birds
mammals
see UV light
amniotic egg
spiny-rayed fins
lungs
molt exoskeleton
vertebrae


Land Plants
Birds
Insects
Reptiles
Mammals
Amphibians
Crustacean
Arachnids
Mollusks
Seaweed
Fishes
Bryozoans
Brachiopod
Vertebrates
Worms
Echinoderms
Sponges
Coelenterates
Protophytes
Protozoans
Protists


trout
humar
bird
insect
amoeba
bacterium
insect
trout
amoeba
bird
bacterium
human
bacterium
amoeba
insect
trout
human
bird

The Geological Time Scale

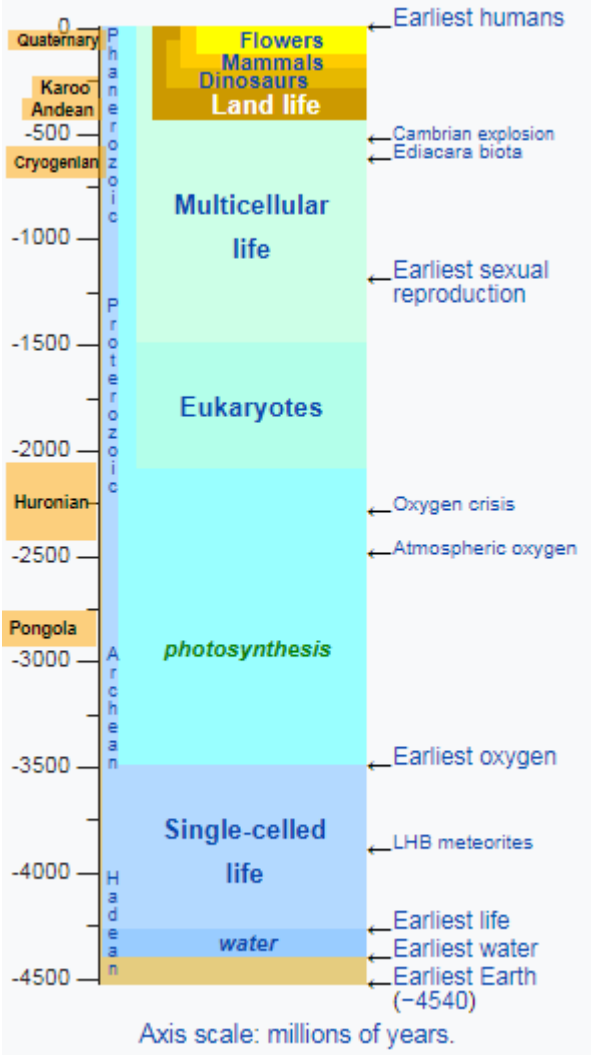
- Enormous time span involved in the evolutionary process
  - First life approx 3.5 billion years ago
- Divided into eras, periods, epochs:

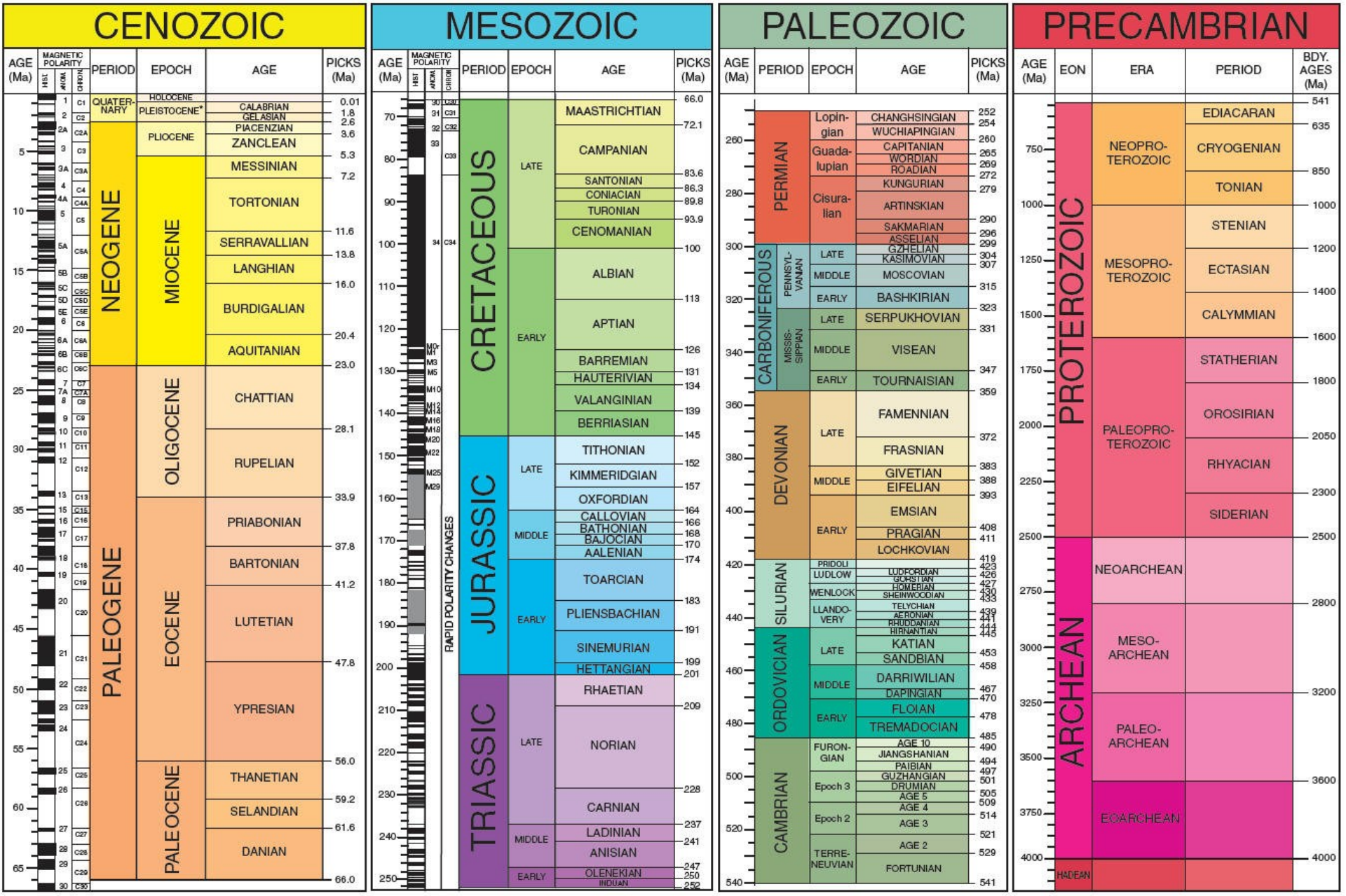
Table 16.2 The geological time scale covering the past 600 million years

Era	Period	Epoch	Millions of years BP
Cainozoic (Cenozoic)	Quaternary	Holocene (Recent)	0.012 (11 700 years)
		Pleistocene	2.6
	Tertiary	Pliocene	5.3
		Miocene	23
		Oligocene	34
		Eocene	56
		Palaeocene	65
Mesozoic	Cretaceous		145
	Jurassic		200
	Triassic		251
Palaeozoic	Permian		299
	Carboniferous		359
	Devonian		416
	Silurian		444
	Ordovician		488
	Cambrian		542
	Ediacaran		635

The Geological Time Scale

- Humans and human ancestors relatively very recent:
  - Cainozoic onwards




CENOZOIC
MESOZOIC
PALEOZOIC
PRECAMBRIAN
AGE (Ma)
MAGNETIC POLARITY
HIST
ANOM
CHRON
PERIOD
EPOCH
AGE
PICKS (Ma)
QUATER-NARY
NEOGENE
PALEOGENE
HOLOCENE
PLEISTOCENE*
PLIOCENE
MIOCENE
OLIGOCENE
EOCENE
PALEOCENE
CALABRIAN
GELASIAN
PIACENZIAN
ZANCLEAN
MESSINIAN
TORTONIAN
SERRAVALLIAN
LANGHIAN
BURDIGALIAN
AQUITANIAN
CHATTIAN
RUPELIAN
PRIABONIAN
BARTONIAN
LUTETIAN
YPRESIAN
THANETIAN
SELANDIAN
DANIAN
0.01
1.8
2.6
3.6
5.3
7.2
11.6
13.8
16.0
20.4
23.0
28.1
33.9
37.8
41.2
47.8
56.0
59.2
61.6
66.0
CRETACEOUS
JURASSIC
TRIASSIC
LATE
EARLY
MIDDLE
MAASTRICHTIAN
CAMPANIAN
SANTONIAN
CONIACIAN
TURONIAN
CENOMANIAN
ALBIAN
APTIAN
BARREMIAN
HAUTERIVIAN
VALANGINIAN
BERRIASIAN
TITHONIAN
KIMMERIDGIAN
OXFORDIAN
CALLOVIAN
BATHONIAN
BAJOCIAN
AALENIAN
TOARCIAN
PLIENSBACHIAN
SINEMURIAN
HETTANGIAN
RHAETIAN
NORIAN
CARNIAN
LADINIAN
ANISIAN
OLENEKIAN
INDUAN
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86.3
89.8
93.9
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RAPID POLARITY CHANGES
PERMIAN
CARBONIFEROUS
PENNSYL-VANIAN
MISSIS-SIPPIAN
DEVONIAN
SILURIAN
ORDOVICIAN
CAMBRIAN
Lopin-gian
Guada-lupian
Cisura-lian
CHANGHSINGIAN
WUCHIAPINGIAN
CAPITANIAN
WORDIAN
ROADIAN
KUNGURIAN
ARTINSKIAN
SAKMARIAN
ASSELIAN
GZHELIAN
KASIMOVIAN
MOSCOVIAN
BASHKIRIAN
SERPUKHOVIAN
VISEAN
TOURNAISIAN
FAMENNIAN
FRASNIAN
GIVETIAN
EIFELIAN
EMSIAN
PRAGIAN
LOCHKOVIAN
PRIDOLI
LUDLOW
WENLOCK
LLANDO-VERY
LUDFORDIAN
GORSTIAN
HOMERIAN
SHEINWOODIAN
TELYCHIAN
AERONIAN
RHUDDANIAN
HIRNANTIAN
KATIAN
SANDBIAN
DARRIWILIAN
DAPINGIAN
FLOIAN
TREMADOCIAN
FURON-GIAN
Epoch 3
Epoch 2
TERRE-NEUVIAN
AGE 10
JIANGSHANIAN
PAIBIAN
GUZHANGIAN
DRUMIAN
AGE 5
AGE 4
AGE 3
AGE 2
FORTUNIAN
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505
509
514
521
529
541
EON
ERA
BDY. AGES (Ma)
PROTEROZOIC
ARCHEAN
HADEAN
NEOPRO-TEROZOIC
MESOPRO-TEROZOIC
PALEOPRO-TEROZOIC
NEOARCHEAN
MESO-ARCHEAN
PALEO-ARCHEAN
EOARCHEAN
EDIACARAN
CRYOGENIAN
TONIAN
STENIAN
ECTASIAN
CALYMMIAN
STATHERIAN
OROSIRIAN
RHYACIAN
SIDERIAN
541
635
850
1000
1200
1400
1600
1800
2050
2300
2500
2800
3200
3600
4000

Problems with the fossil record

- Incomplete: few fossils formed as specific conditions required:
  - Quick burial
  - Presence of hard body parts, or imprint
  - Absence of decay organisms
  - Long period of stability
  - Correct environmental conditions eg soil pH
- Only some fossils formed get found
- Dating material can be problematic in some cases
- Fossils often incomplete